Computer Networks

Home Assignment-3

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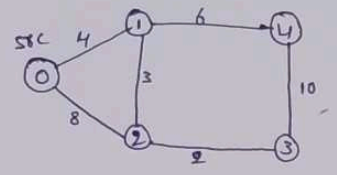
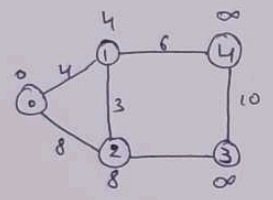
1) Dijkstra's shortest-path algorithm



1



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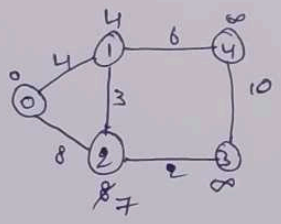
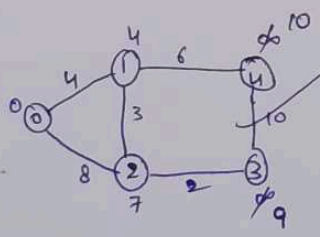


3



the shortest paths from 0 are

$0 \to 1 = 4$

$0 \to 2 = 7$

$0 \to 3 = 9$

$0 \to 4 = 10$

2) Distance vector vs Link State Routing

Distance vector Routing :-

- Routers share routing tables only with neighbours.
- Convergence time is slow.
- Low overhead.
- Example : RIP.

Link State routing :-

- Routers share link state information with all routers.
- Convergence time is ~~slow~~ fast.
- Higher overhead due to flooding.
- Example : OSPF

Count to infinity :-

In Distance vector routing, when a route fails, routers continuously increase the hop count for that destination until it reaches infinity, causing slow convergence.

Eg:- In network A-B-C, if link A-B fails, B and C keep updating routes to A through each other, increasing hop count.

3) ① users enter a domain name in the browser.
② Request goes to the local DNS resolver.
③ Resolver queries the Root DNS server.
④ Root server returns the TLD server (.com).
⑤ Resolver queries the TLD server.
⑥ TLD returns the Authoritative DNS server.
⑦ Authoritative server returns the IP address.
⑧ Browser connects to the web server using that IP.

4) TCP connection

ⓐ Three - way handshake
① Client -> Server : SYN (Seq = 1000)

② Server → Client : SYN + ACK (Seq = 5000, Ack = 1001)
③ Client → Server : ACK (Seq = 1001, Ack = 5001)
Connection established.

ⓑ Data transfer

client sends 800 bytes → Next Seq = 1801, Server Ack = 1801.
server sends 500 bytes → Next Seq = 5501, Client Ack = 5501

5) TCP Header Structure

TCP header contains fields used for reliable data transmission.

Main fields:-

- Source port - sender application
- Destination Port - reciever application
- Sequence Number - order of data
- Acknowledgement Number - confirms recieved data.
- Header Length - size of header.
- Flags - SYN, ACK, FIN control connection.
- Window Size - flow control.
- Checksum - error detection
- Urgent Pointer & Options - additional control information.

6) Leaky Bucket Algorithm

Bucket size = 10 packets
Output rate = 2 packets / sec
If 12 packets arrive at once:

- Bucket stores 10 packets.
- 2 packets are dropped.

Packets leave bucket:

- 2 packets per second
- Bucket becomes empty in 5 seconds.

7) Open Loop vs Closed Loop Congestion Control

Open Loop	Closed Loop
① No network feedback	① uses feedback from network
② prevents congestion	② Detects and controls congestion
③ Simple mechanism	③ More complex
④ Eg:- Traffic shaping	④ Example:- TCP congestion control

8) ACL Configuration

Type: Standard ACL

Configuration

```
access-list 10 deny 192.5.6.2
access-list 10 deny 192.5.6.3
access-list 10 permit any
```

Apply on interface

```
interface g0/0
ip access-group 10 in
```

Direction: Inbound